

Class 04 — Homework

Bong Study Hub — Foundation Batch 2026 · Class 04: How Computers Represent Information

From Information → Bits → Binary

Estimated time: 25–40 minutes **Total:** 30 marks · **Optional Challenge:** up to 3 bonus marks **Tools needed:** pen and paper, or a simple document — nothing else **Prerequisite:** Class 04. No programming, no electronics, and no binary arithmetic required.

What This Assignment Is About

In class, we asked how a computer can represent information as different as a number, a word, a photograph, or a song — using nothing but combinations of two-state units called **bits**. We saw that one bit gives two possibilities, that more bits give more combinations, and that a bit pattern only means something once we agree on how to interpret it.

This assignment asks you to **reason with these ideas, not recite them**. There is no binary arithmetic, no conversion between binary and decimal, and no single "correct sentence" to memorize — if your reasoning makes sense, your answer is good, even if it doesn't use the exact words from class.

Instructions

- **Answer in your own words.** The goal is to show that you understand the idea, not to reproduce definitions exactly.
- Several questions are open-ended — there may be more than one reasonable answer.
- **No programming, electronics, or binary-arithmetic knowledge is required.**
- You may refer to your **Student Notes** while answering.
- If you're unsure, **explain your reasoning anyway** rather than leaving a blank — partial thinking is worth more than an empty space.

Section A — Core Concepts

Question 1 — Information Needs Representation (2 marks)

In your own words, explain **why a computer needs a representation of information before it can store or process it**.

Question 2 — Two-State Systems (2 marks)

A system has two clearly distinguishable states — for example, a light switch that is either off or on.

Explain why a system like this is useful for digital representation.

Question 3 — What Is a Bit? (2 marks)

In your own words, **explain what a bit represents.**

Question 4 — One Bit (2 marks)

How many possible states can one bit represent? Briefly explain why.

Question 5 — Different Kinds of Information (2 marks)

Name at least three different kinds of information (besides numbers)
that can be represented using bits.

Section B — Reasoning & Representation

Question 6 — Two Bits (3 marks)

Suppose you have **two bits** to work with.

List every possible combination, then state how many combinations there are in total and how you know you've found them all.

#	Combination
1	
2	
3	
4	

Question 7 — Predicting Growth (3 marks)

We saw this pattern in class:

1 bit → 2 combinations
2 bits → 4 combinations
3 bits → 8 combinations

Predict how many combinations 4 bits would give, and explain your reasoning. You do not need a formula — reason from the pattern.

Question 8 — Representation ≠ Information (3 marks)

In Class 03, we said that "**80% full**" represents a crowded canteen — it is not the canteen itself.

Explain the similarity between this idea and a binary bit pattern representing information.

Question 9 — Where Does Meaning Come From? (3 marks)

Imagine a hypothetical traffic-signal system that uses 2 bits like this:

Bits	Meaning
00	Stop
01	Slow down
10	Speed up
11	Go

Do these meanings come from the bits themselves? Explain your answer.

Question 10 — Create Your Own Representation (3 marks)

You have **two bits** and need to represent **four different states** of something you choose — for example, four moods, four weather conditions, or four traffic signals.

Create your own mapping:

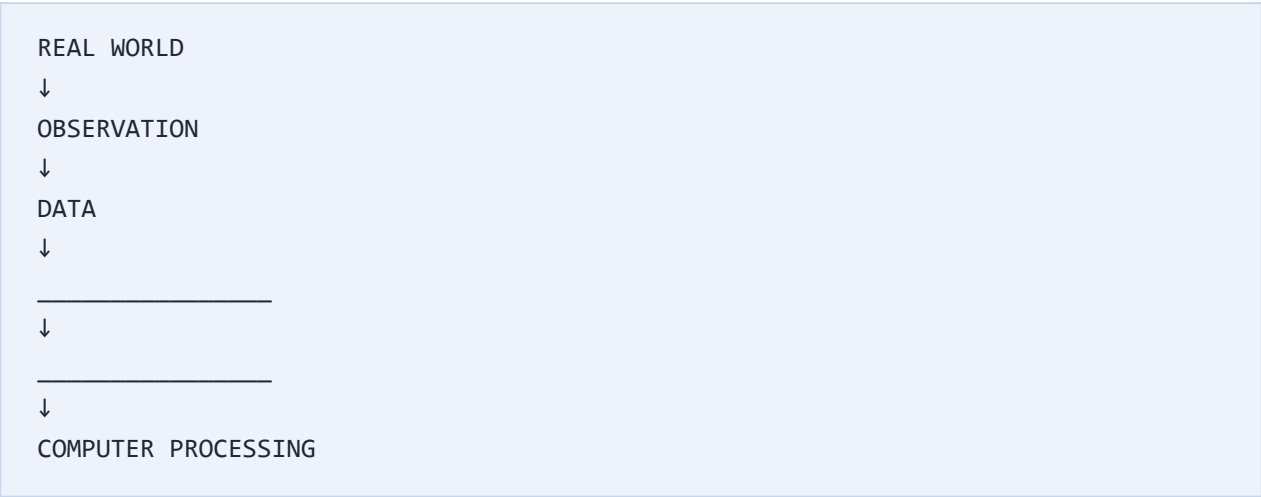
Bits	Your Meaning
00	
01	
10	
11	

Then briefly **explain why your system works**.

Section C — Class Connection & Synthesis

Question 11 — Connecting Class 03 and Class 04 (2 marks)

Fill in the two blanks below:



Briefly explain, in one sentence, what Class 04 added to Class 03's chain.

Question 12 — Synthesis (3 marks)

Why is it powerful that the same basic building block — bits — can be used to represent numbers, text, images, sound, and video? Answer in 2–4 sentences.

Optional Challenge — Bonus (up to 3 marks)

This section is entirely optional and will not affect your base grade.

Imagine a system with 3 bits.

- List the possible combinations.
- Invent a meaning for each combination.
- Explain why your meanings are not inherent in the bit patterns.

Bits	Your Meaning

Submission Format

Choose whichever is easiest for you:

- **Option A:** Handwrite your answers, then photograph or scan them.
- **Option B:** Type your answers in a simple document.
- **Option C:** Submit as a simple PDF.

Please label each answer with its question number.

Before You Submit — Self-Check

- ☐ I answered all 12 questions in my own words.
- ☐ Where unsure, I explained my reasoning instead of leaving a blank.
- ☐ I did not use binary arithmetic, conversions, or code anywhere.
- ☐ I reviewed my work before submitting.